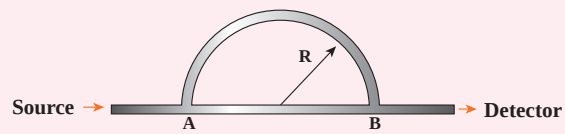
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13. Write short notes on intensity and loudness.
14. Explain how overtones are produced in a
    (a) Closed organ pipe
    (b) Open organ pipe
15. How will you determine the velocity of sound using resonance air column apparatus?
16. What is meant by Doppler effect?. Discuss the following cases
    **(1)** Source in motion and Observer at rest
        (a) Source moves towards observer
        (b) Source moves away from the observer
    **(2)** Observer in motion and Source at rest.
        (a) Observer moves towards Source
        (b) Observer resides away from the Source
    **(3)** **B**oth are in motion
        **(a)** Source and Observer approach each other
        (b) Source and Observer resides from each other
        (c) Source chases Observer
        (d) Observer chases Source

## IV. Numerical Problems

**1.**The speed of a wave in a certain medium is 900 m/s. If 3000 waves passes over a certain point of the medium in 2 minutes, then compute its wavelength?. **Answer** : $\lambda = 36$ m

2. Consider a mixture of 2 mol of helium and 4 mol of oxygen. Compute the speed of sound in this gas mixture at 300 K. **Answer** : 400.9 $ms^{-1}$
3. A ship in a sea sends SONAR waves straight down into the seawater from the bottom of the ship. The signal reflects from the deep bottom bed rock and returns to the ship after 3.5 s. After the ship moves to 100 km it sends another signal which returns back after 2s. Calculate the depth of the sea in each case and also compute the difference in height between two cases. **Answer** : $\Delta d = 1149.75$ m
4. A sound wave is transmitted into a tube as shown in figure. The sound wave splits into two waves at the point A which recombine at point B. Let R be the radius of the semi-circle which is varied until the first minimum. Calculate the radius of the semi-circle if the wavelength of the sound is 50.0 m. **Answer** : $R = 21.9\ m$




5. N tuning forks are arranged in order of increasing frequency and any two successive tuning forks give n beats per second when sounded together. If the last fork gives double the frequency of the first (called as octave), Show that the frequency of the first tuning fork is $f = (N-1)n$.
6. Let the source propagate a sound wave whose intensity at a point (initially) be $I$. Suppose we consider a case when the amplitude of the sound wave is